

蓝桥杯国赛JavaB组2021

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本文记录了2021蓝桥杯国赛JavaB组的部分题解

1. 整数范围

```
import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);

    public static void main(String[] args) {
        System.out.println((int)Math.pow(2, 8) - 1);
    }
}
```

2. 纯质数

```
import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);
    static int N = 20210610, n, cnt;
    static boolean[] st = new boolean[N];
    static int[] p = new int[N];
    public static void get_p(int num) {
        for(int i = 2; i <= num; i++) {
            if (!st[i]) p[cnt++] = i;
            for(int j = 0; p[j] <= num / i; j++) {
                st[p[j] * i] = true;
                if (i % p[j] == 0) break;
            }
        }
    }

    public static void main(String[] args) {
        n = 20210605;
        int[] a = new int[] {2,3,5,7};
        get_p(n);
        int res = 0;
        for(int i = 0; i < cnt; i++) {
            int x = p[i];
            boolean flag = true;
            while(x > 0) {
                int t = x % 10;
                if (t != 2 && t != 3 && t != 5 && t != 7) {

```

```

        flag = false;
        break;
    }
    x /= 10;
}
if (x > 0 && x != 2 && x != 3 && x != 5 && x != 7) flag = false;
if (flag) res++;
}
System.out.println(res);
}
}

```

3. 完全日期

```

import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);

    public static void main(String[] args) {
        int cnt = 0;
        for(int i = 2001; i <= 2021; i++) {
            for(int j = 1; j <= 12; j++) {
                int k = getDay(i, j);
                for(int t = 1; t <= k; t++) {
                    int sum = getSum(i) + getSum(j) + getSum(t);
                    if (Math.sqrt(sum) == (int)Math.sqrt(sum)) {
                        cnt++;
                    }
                }
            }
        }
        System.out.println(cnt);
    }

    public static int getDay(int y, int m) {
        int[] day = {31,0,31,30,31,30,31,31,30,31,30,31};
        if (m != 2) {
            return day[m-1];
        }
        if ((y % 4 == 0 && y % 100 != 0) || (y % 400 == 0)) {
            return 29;
        }
        return 28;
    }

    public static int getSum(int num) {
        int sum = 0;
        while(num != 0) {
            sum += num % 10;
            num /= 10;
        }
        return sum;
    }
}

```

4. 最小权值

```
import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);
    static int N = 2030;
    static long[] dp = new long[N];
    public static void main(String[] args) {
        Arrays.fill(dp, 0x3f3f3f3f);
        dp[0] = 0;
        dp[1] = 1;
        for(int i = 2; i <= 2021; i++) {
            dp[i] = 1 + 2 * dp[i-1];
            for(int j = 1; j < i; j++) {
                dp[i] = Math.min(dp[i], 1 + 2 * dp[j] + 3 * dp[i - 1 - j] + j * j * (i - 1 - j));
            }
        }

        System.out.println(dp[2021]);
    }
}
```

5. 大写

```
import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);

    public static void main(String[] args) {
        String str = sc.next();
        System.out.println(str.toUpperCase());
    }
}
```

6. 123

```
// 过70%样例
import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);
    public static long get(int x, int y) {
        long res = 0;
        for(int i = x; i <= y; i++) {
            res += (1 + i) * i / 2;
        }
        return res;
    }
}
```

```

public static int find(int d) {
    for(int i = 1; i <= d; i++) {
        if ((i + 1) * i / 2 >= d) {
            return i;
        }
    }
    return 0;
}

public static void main(String[] args) {
    int t = sc.nextInt();
    while(t-->0) {
        long res = 0;
        int l = sc.nextInt();
        int r = sc.nextInt();
        int x = find(l);
        int y = find(r);
        if (x != y) {
            if (x+1 <= y-1) res += get(x+1, y-1);
            l -= x * (x - 1) / 2;
            r -= y * (y - 1) / 2;
            res += (1 + x) * (x - l + 1) / 2;
            res += (1 + r) * r / 2;
        }
        else {
            l -= x * (x - 1) / 2;
            r -= y * (y - 1) / 2;
            res += (1 + r) * (r - l + 1) / 2;
        }
        System.out.println(res);
    }
}
}

```

7. 和与乘积

8. 巧克力

```

import java.util.*;

public class Main {
    static final Scanner sc = new Scanner(System.in);
    static int N = 100010, n, x;
    public static void main(String[] args) {
        x = sc.nextInt();
        n = sc.nextInt();
        choc[] chocs = new choc[n];
        for(int i = 0; i < n; i++) {
            chocs[i] = new choc(sc.nextInt(), sc.nextInt(), sc.nextInt());
        }
        // 优先保证保质期
        Arrays.sort(chocs, (o1, o2) -> {
            if (o1.b != o2.b) {
                return o2.b - o1.b;
            } else {
                return o1.a - o2.a;
            }
        });
    }
}

```

```

    }
});
// 优先队列根据价格排序
PriorityQueue<choc> pq = new PriorityQueue<>((o1,o2) -> o1.a - o2.a);
long cost = 0; // 记录总花费
int pos = 0; // 记录巧克力的索引
for(int i = x; i >= 1; i--) {
    while(pos < n && chocs[pos].b >= i) {
        pq.add(chocs[pos]);
        pos++;
    }

    if (pq.isEmpty()) {
        System.out.println("-1");
        return;
    }

    choc t = pq.poll();
    cost += t.a;
    t.c--;
    if (t.c > 0) {
        pq.add(t);
    }
}

System.out.println(cost);
}
}

class choc{
    int a;
    int b;
    int c;
    public choc(int a, int b, int c) {
        this.a = a;
        this.b = b;
        this.c = c;
    }
}
}

```